

# High-Dimensional Arrhythmia Classification from Raw ECG Time Series

Data understanding, preprocessing, model tuning, evaluation, and RMT diagnostics

ECG RMT Analysis

May 2026

# Task Definition

## Goal

Build a reproducible binary ECG screening pipeline from PTB-XL raw WFDB waveforms and metadata, then explain how fitted models behave in the high-dimensional feature regime.

- ▶ Input: ten-second, twelve-lead ECG waveform records plus patient and acquisition metadata.
- ▶ Target: normal versus diagnostic abnormality from SCP diagnostic superclasses.
- ▶ Data split: official PTB-XL folds; train, validation, and held-out test are kept separate.
- ▶ Output: trained model comparison, final metrics, paper, presentation, and MLflow logs.

# EDA: Dataset and Label Construction

## PTB-XL inputs

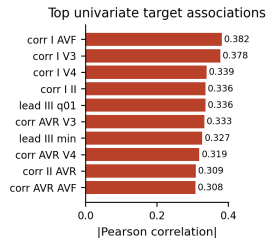
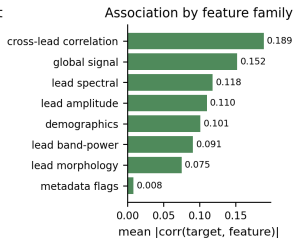
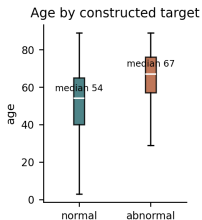
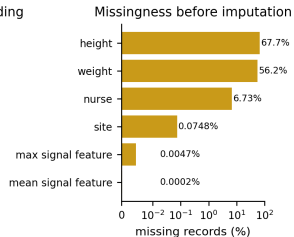
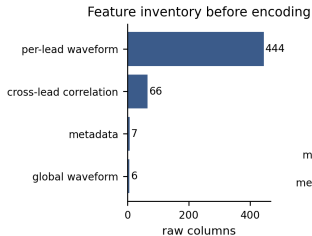
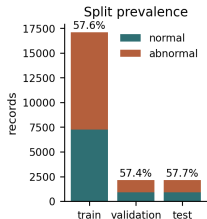
- ▶ 21,799 metadata rows.
- ▶ WFDB records at 100 Hz and 500 Hz.
- ▶ Twelve simultaneous ECG leads.
- ▶ SCP-ECG code dictionary for diagnostic mapping.

## Constructed supervised table

| Quantity               | Count  |
|------------------------|--------|
| Labeled records        | 21,388 |
| Normal                 | 9,069  |
| Diagnostic abnormality | 12,319 |
| Unlabeled or unmapped  | 411    |

The binary target is a screening abstraction: a record is positive if at least one mapped diagnostic superclass is not NORM.

# EDA: What We Found Before Modeling



Rows: 21,388; raw feature map: 523 columns; preprocessed design matrix: 594 columns after one-hot expansion.

# EDA: Key Data Characteristics

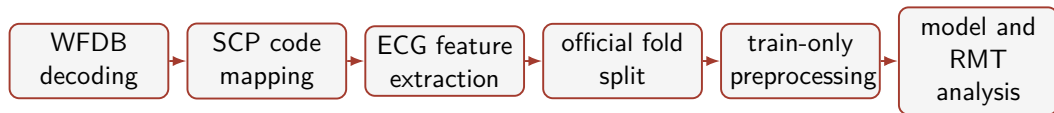
## Class and fold structure

- ▶ Train: 17,084 records.
- ▶ Validation: 2,146 records.
- ▶ Test: 2,158 records.
- ▶ Abnormal class prevalence is near 58% in each split.

## Quality observations

- ▶ Height and weight are highly incomplete, so median imputation is required.
- ▶ Waveform-derived features are almost complete after decoding.
- ▶ Age differs between target groups, so demographics are retained but regularized.
- ▶ Cross-lead correlations show strong univariate association with the target.

# Preprocessing Pipeline



## Leakage control

Imputation, scaling, and one-hot expansion are fit on training records only and then applied to validation/test records through the same pipeline object.

# Preprocessing and Feature Handling

| Numeric features                                                                         | Categorical metadata                                                                       | Feature dimensions                                                      |
|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| median imputation followed by standard scaling for linear models and covariance analysis | most-frequent imputation and one-hot encoding with unknown categories ignored at test time | 523 raw modeling columns become 594 preprocessed columns after encoding |

$$z_i = \phi(\text{waveform}_i, \text{metadata}_i) \in \mathbb{R}^{523}, \quad x_i = g(z_i) \in \mathbb{R}^{594}.$$

# Feature Engineering Choices

| Feature group                  | Reason for inclusion                                                        |
|--------------------------------|-----------------------------------------------------------------------------|
| Lead amplitude statistics      | Capture waveform scale, spread, asymmetry, and robust quantiles.            |
| Derivative and energy features | Represent local morphology and signal roughness without hand-labeled beats. |
| FFT and band-power features    | Add spectral information for rhythm and noise-sensitive structure.          |
| Cross-lead correlations        | Encode multi-lead coordination and axis-like relationships.                 |
| Metadata                       | Retain clinically relevant acquisition and demographic context.             |

## Design choice

The project uses interpretable engineered features instead of a deep ECG network so that preprocessing, feature families, and RMT spectra are directly inspectable.



## Model Families and Why They Were Chosen

| Model                       | Purpose in the comparison                                                        |
|-----------------------------|----------------------------------------------------------------------------------|
| Logistic regression         | Regularized linear baseline for high-dimensional tabular features.               |
| SGD linear                  | Fast linear margin model with tunable $L_1/L_2$ -style regularization.           |
| Random forest               | Bagged nonlinear ensemble with variance reduction.                               |
| Extra trees                 | More randomized tree ensemble to test sensitivity to split randomness.           |
| Histogram gradient boosting | Strong CPU boosting baseline with explicit $L_2$ regularization.                 |
| XGBoost                     | Regularized boosted trees with $L_1$ , $L_2$ , subsampling, and column sampling. |

# Training and Tuning Protocol

## Methodology

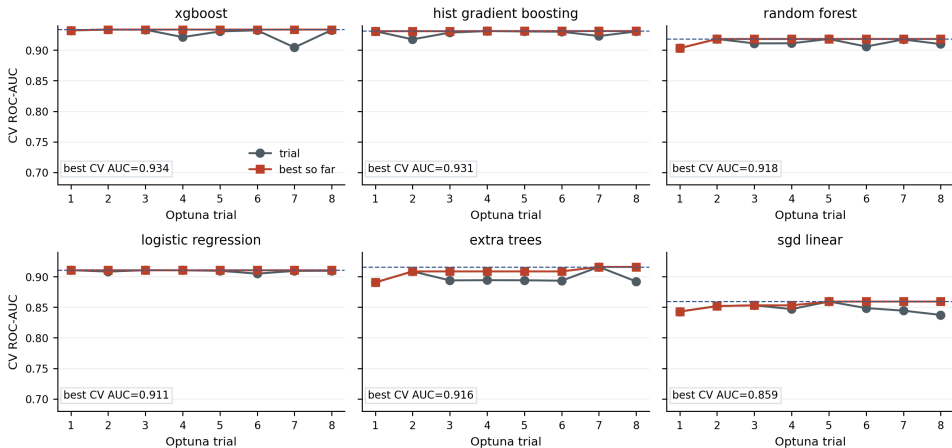
- ▶ Train fold is used for fitting and cross-validation.
- ▶ Optuna TPE search optimizes CV ROC-AUC.
- ▶ Validation fold checks model-family selection.
- ▶ Test fold is held out for final reported metrics.

## Tuned hyperparameters

- ▶ Linear:  $C$ ,  $\alpha$ , penalty,  $L_1$  ratio.
- ▶ Forests: trees, depth, leaf size, feature sampling.
- ▶ Boosting: learning rate, iterations, leaves,  $L_2$ .
- ▶ XGBoost: depth, trees, sampling,  $L_1$ ,  $L_2$ .

# Systematic Hyperparameter Tuning

## Systematic Hyperparameter Tuning



# Evaluation Metrics

## Why multiple metrics?

The target is moderately imbalanced, so accuracy alone is not enough. We report ROC-AUC, F1, balanced accuracy, precision, recall, and confusion matrices.

| Metric            | Interpretation                                           |
|-------------------|----------------------------------------------------------|
| ROC-AUC           | Ranking quality across thresholds.                       |
| F1                | Precision-recall balance at the operating threshold.     |
| Balanced accuracy | Average of sensitivity and specificity.                  |
| Confusion matrix  | Error type: missed abnormal records versus false alarms. |

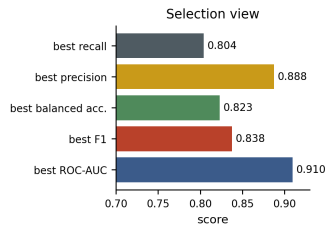
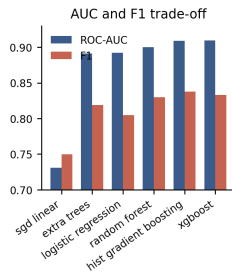
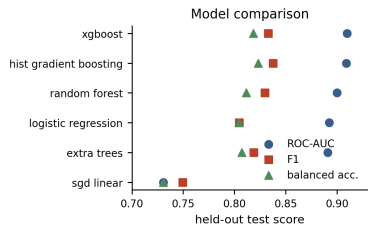
## Final Held-Out Test Metrics

| Model                   | Acc.   | Bal. acc. | Prec.  | Recall | F1     | ROC-AUC |
|-------------------------|--------|-----------|--------|--------|--------|---------|
| Logistic regression     | 0.7938 | 0.8044    | 0.8877 | 0.7360 | 0.8047 | 0.8924  |
| SGD linear              | 0.7271 | 0.7307    | 0.7973 | 0.7071 | 0.7495 | 0.7307  |
| Random forest           | 0.8100 | 0.8115    | 0.8597 | 0.8018 | 0.8297 | 0.9001  |
| Extra trees             | 0.8021 | 0.8072    | 0.8686 | 0.7745 | 0.8188 | 0.8909  |
| Hist. gradient boosting | 0.8202 | 0.8231    | 0.8743 | 0.8042 | 0.8378 | 0.9089  |
| XGBoost                 | 0.8151 | 0.8183    | 0.8712 | 0.7978 | 0.8328 | 0.9097  |

### Main comparison

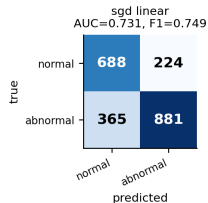
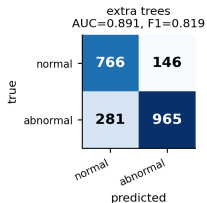
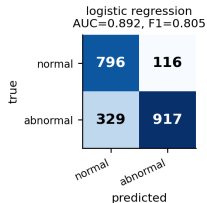
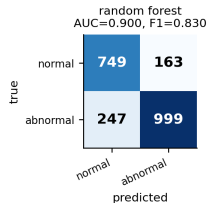
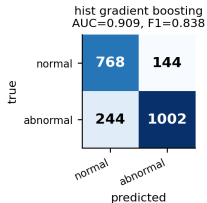
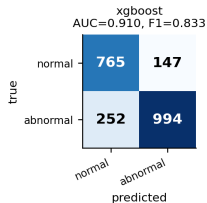
XGBoost obtains the highest test ROC-AUC, while histogram gradient boosting obtains the highest test F1 and balanced accuracy.

# Evaluation Visualization



# Confusion Matrices

## Held-Out Test Confusion Matrices



Counts are reconstructed from the reported test precision/recall and official test split class counts: normal=912, abnormal=1,246.

# Random Matrix Theory Diagnostic

## Marchenko–Pastur reference

For standardized white-noise features with aspect ratio  $\gamma = p/n$ , sample-covariance eigenvalues concentrate on

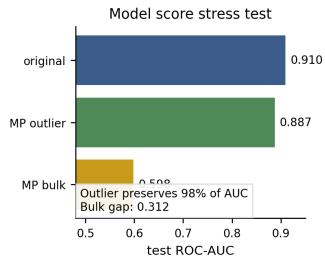
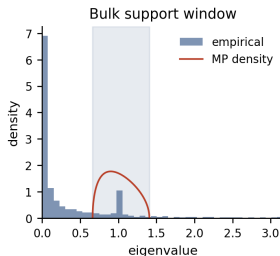
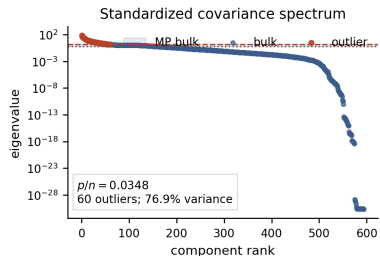
$$\lambda_{\pm} = (1 \pm \sqrt{\gamma})^2.$$

- ▶ Eigenvalues inside  $[\lambda_-, \lambda_+]$  form the empirical bulk.
- ▶ Eigenvalues above  $\lambda_+$  are structured outliers.
- ▶ The experiment checks whether model scores survive if the data are projected onto outlier or bulk subspaces.



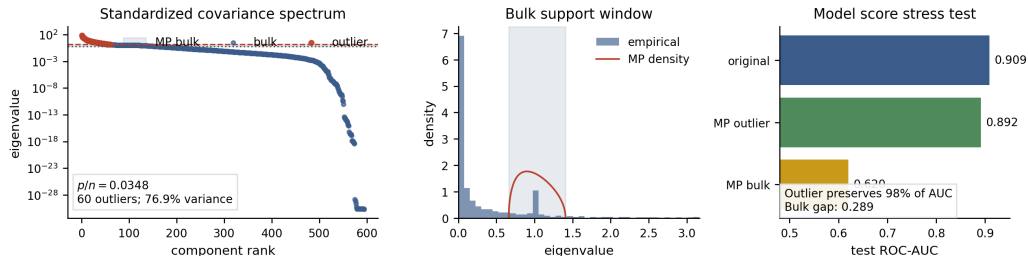
# RMT Model Behavior: XGBoost

RMT Model Behavior: xgboost



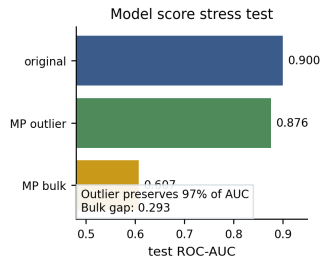
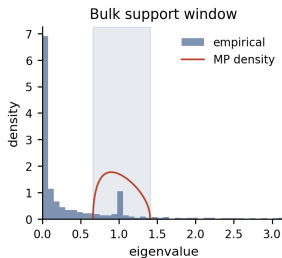
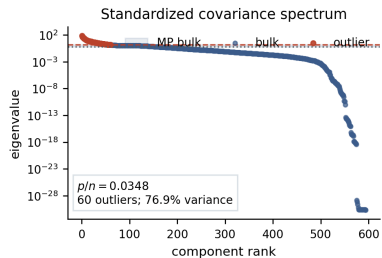
# RMT Model Behavior: Histogram Gradient Boosting

RMT Model Behavior: hist gradient boosting



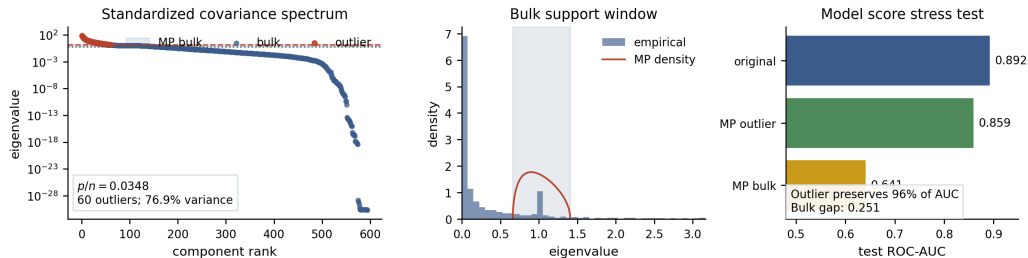
# RMT Model Behavior: Random Forest

RMT Model Behavior: random forest



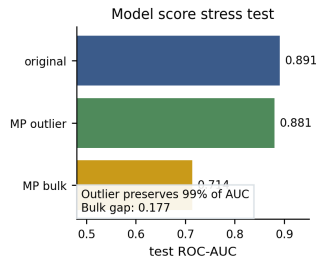
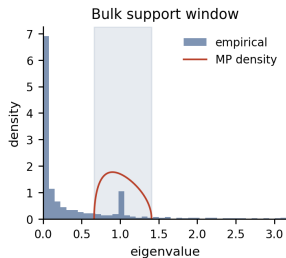
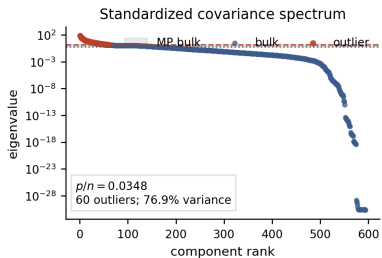
# RMT Model Behavior: Logistic Regression

RMT Model Behavior: logistic regression



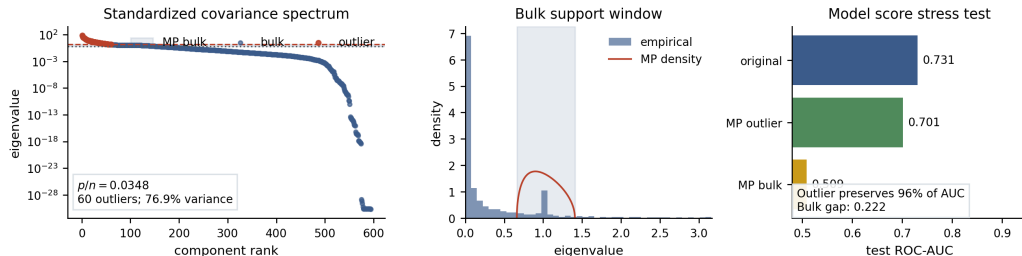
# RMT Model Behavior: Extra Trees

RMT Model Behavior: extra trees

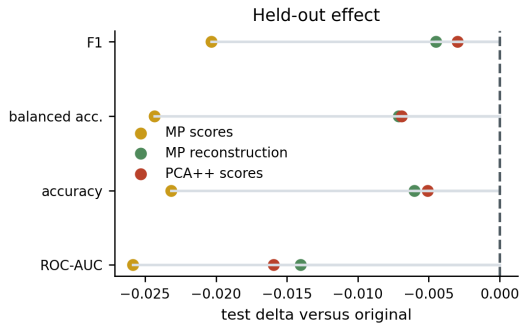
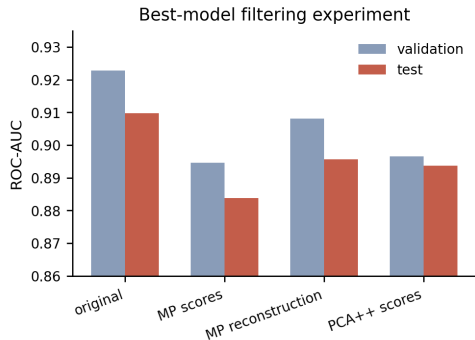


# RMT Model Behavior: SGD Linear

RMT Model Behavior: sgd linear



# Filtering Experiment



# Interpretation

- ▶ The covariance spectrum has  $p/n = 0.0348$ , MP support  $[0.662, 1.408]$ , and 60 outlier components.
- ▶ Those outliers explain 76.9% of standardized variance, so the feature map has a compact dominant subspace.
- ▶ MP-outlier reconstructions preserve much of the predictive signal.
- ▶ MP-bulk reconstructions are weaker but not empty; hard filtering removes weak useful information.
- ▶ This explains why RMT is useful as model analysis here, but not as a direct accuracy booster.



Thank you